

## Activity 1 – RSA Manual Calculations

### 1. Choose primes

$$p=2, q=3 \quad p=2, \backslash; q=3 \quad p=2, q=3$$

### 2. Compute $n$

$$n=p \times q=2 \times 3=6 \quad n=p \times q=2 \times 3=6 \quad n=p \times q=2 \times 3=6$$

### 3. Compute Euler's totient $\phi(n)$

$$\phi(n)=(p-1)(q-1)=(2-1)(3-1)=1 \times 2=2 \quad \varphi(n)=(p-1)(q-1)=(2-1)(3-1)=1 \times 2=2 \\ 2\phi(n)=(p-1)(q-1)=(2-1)(3-1)=1 \times 2=2$$

### 4. Choose public exponent $e$ such that $1 < e < \phi(n)$ and $\gcd(e, \phi(n)) = 1$

- Try  $e=3$ :  $\gcd(3, 2) = 1$
- So  $e=3$ .

### 5. Compute private exponent $d$ as the modular inverse of $e$ mod $\phi(n)$ :

$$d \equiv e^{-1} \pmod{\phi(n)} \quad d \equiv e^{-1} \pmod{\phi(n)} \quad d \equiv e^{-1} \pmod{\phi(n)}$$

We need  $d$  such that

$$d \times e \equiv 1 \pmod{2} \quad d \times e \equiv 1 \pmod{2} \quad d \times e \equiv 1 \pmod{2}$$

Since  $3 \equiv 1 \pmod{2}$ , we solve

$$d \times 1 \equiv 1 \pmod{2} \Rightarrow d \equiv 1 \pmod{2} \Rightarrow d=1 \quad d \times 1 \equiv 1 \pmod{2} \Rightarrow d \equiv 1 \pmod{2} \Rightarrow d=1$$

### 6. Keys

- Public key:  $(n=6, e=3)$
- Private key:  $(n=6, d=1)$

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## Task 2 – Encrypt MMM

Let  $M=5$ .

$$C = M \pmod{n} = 5 \pmod{6} = 5 \quad C = M \pmod{n} = 5 \pmod{6} = 5 \quad C = M \pmod{n} = 5 \pmod{6} = 5$$

## Task 3 – Decrypt CCC

$$M' = C \pmod{n} = 5 \pmod{6} = 5 \quad M' = C \pmod{n} = 5 \pmod{6} = 5 \quad M' = C \pmod{n} = 5 \pmod{6} = 5$$

Recovered plaintext  $M'=5$ , matching the original.

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## Activity 2 – Diffie–Hellman Manual Calculations

- Partner's name: Prasad Polishetty
- Public parameters:  $p=7$ ,  $g=3$

### 1. Your private key

$a=4$  (keep secret)

### 2. Your public key

$A = g^a \bmod p = 3^4 \bmod 7 = 81 \bmod 7 = 4$   
 $A = g^a \bmod p = 3^4 \bmod 7 = 81 \bmod 7 = 4$

### 3. (Simulated) Partner's private key

$b=5$  (Prasad's secret—not logged)  
 $b=5$  (Prasad's secret—not logged)

Partner's public key

$B = g^b \bmod p = 3^5 \bmod 7 = 243 \bmod 7 = 5$   
 $B = g^b \bmod p = 3^5 \bmod 7 = 243 \bmod 7 = 5$

### 4. Compute shared secret

- You compute

$s = B^a \bmod p = 5^4 \bmod 7 = 625 \bmod 7 = 2$   
 $s = B^a \bmod p = 5^4 \bmod 7 = 625 \bmod 7 = 2$

- Prasad computes

$s = A^b \bmod p = 4^5 \bmod 7 = 1024 \bmod 7 = 2$   
 $s = A^b \bmod p = 4^5 \bmod 7 = 1024 \bmod 7 = 2$

Both arrive at  $s=2$